

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	AI and Machine Learning Project
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	Your GitHub Repository: <a href="https://github.com/<your-username>/<your-repository>">https://github.com/<your-username>/<your-repository> Live Project: https://yojana4060.pythonanywhere.com/
Programming Language(s)	Python, HTML5, CSS3
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Joblib

Key Features Implemented	• User-friendly web interface for applicant details. • Machine learning model using Random Forest Classifier. • Data preprocessing and feature scaling. • Real-time credit card approval prediction. • Displays approval/rejection result with prediction accuracy. • Model deployment using PythonAnywhere.
Pending / Incomplete Features	• User authentication and login. • Database integration for storing applications. • Email/SMS notification after prediction. • Admin dashboard and analytics.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install required packages using <code>pip install -r requirements.txt</code>. 3. Run <code>train_model.py</code> (if retraining is needed). 4. Start the Flask application using <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000</code> locally or access the deployed application at <code>https://yojana4060.pythonanywhere.com/</code>.



Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- ☐ The project uses a **Random Forest Classifier** to predict credit card approval based on applicant information.
- ☐ The application is developed using **Python and Flask** with a responsive **HTML/CSS** frontend.
- ☐ The trained model is stored using **Joblib** (model.pkl) and loaded during prediction.
- ☐ The project is deployed on **PythonAnywhere** and managed through **GitHub** for version control.
- ☐ The solution is modular, maintainable, and can be extended with database integration, authentication, and additional machine learning models in future versions.